U2 is swapped with
BUCK module

24Vin
J1
F1
Fuse
+24V
U2
LM7805_TO220
C1
470uF/35v
C4
100n
C6
100n
+5V
5Vout
J9
5V GND
J13
GND
J11
24V out
J14
24Vout
U1
LM7812_TO220
C2
100n
C5
100n
+12V
12Vout
J10
12V GND
J12
GND
L1
SERVO0
+5V
C8
470uF/35v
C9
100n
J4
S0 5v GND
GND
L2
SERVO1
+5V
C15
470uF/35v
C16
100n
J15
S1 5v GND
GND
L3
SERVO2
+5V
C17
470uF/35v
C18
100n
J16
S2 5v GND
GND
+5VD
R1
10k
Digital0
C7
100n
J5
D0 GND
GND
+5VD
R6
10k
Digital1
C29
100n
J21
D1 GND
GND
15.91kHz f cutoff
RV1
+5VD
1k
Analog1
100
R3
9k53
C10
100n
R2
J6
P A1 GND
GND
15.91kHz f cutoff
RV2
+5VD
1k
Analog0
100
R5
9k53
C28
100n
R4
J20
P A0 GND
GND
+5VD
C12
100n
GND
U3
PCF8574AP
SCL
SDA
Addr: 0x20
STEP3
DRV8825
C13
100n
C14
470uF/35v
+24V
GND
J8
M3 A2 A1 B1 B2
GND
+5VD
C19
100n
GND
U4
PCF8574AP
SCL
SDA
Addr: 0x21
STEP2
DRV8825
A3
C20
100n
C21
470uF/35v
+24V
GND
J17
M2 A2 A1 B1 B2
GND
+5VD
C22
100n
GND
U5
PCF8574AP
SCL
SDA
Addr: 0x22
STEP1
DRV8825
A4
C23
100n
C24
470uF/35v
+24V
GND
J18
M1 A2 A1 B1 B2
GND
+5VD
C25
100n
GND
U6
PCF8574AP
SCL
SDA
Addr: 0x23
STEP0
DRV8825
A5
C26
100n
C27
470uF/35v
+24V
GND
J19
M0 A2 A1 B1 B2
GND
Analog2
Analog3
Analog6
Analog7
+5VD
C11
100n
GND
J7
GND 5v A7 A6 A3 A2
Analog IO:

analog0 : lpf 16kHz, used as absolute position sens.
analog1 : lpf 16kHz, used as absolute position sens.
analog2 : directly routed to ardu
analog3 : directly routed to ardu
(4,5 dont exist)
analog6 : directly routed to ardu
analog7 : directly routed to ardu
+12V
C3
100n
GND
+5VD
Arduino_Nano_Every
RESET
AREF
Analog0
Analog1
Analog2
Analog3
SDA
SCL
Analog6
Analog7
STEP0
STEP3
STEP2
STEP1
SERVO2
SERVO1
SERVO0
Digital0
Digital1
J3
GND RX TX
J22
GND SCK MIS MOS
GND
SDA
SCL
SDA SCL GND
J2
GND
Board I/O cheatsheet:

Inputs:

J1 : power 24V in

J20 analog0 input
J6 analog1 input
J7 analog 2,3,6,7 + 5V

Outputs:

J4 servo0 : servo power + signal
J15 servo1 : servo power + signal
J16 servo2 : servo power + signal

J11 : 24V out
J9 : 5V out
J10 : 12 vout

Bidirectional:

J2 — I2C (shared with stepper mux's)
J5 digital0 : general purpose
J21 digial1 : general purpose
H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
H5 MountingHole_Pad
H6 MountingHole_Pad
H7 MountingHole_Pad
H8 MountingHole_Pad
GND

Sheet: /
File: stepperpcb.kicad_sch

Title:

Size: A3 | Date: | Rev:
KiCad E.D.A. 9.0.0 | Id: 1/1